



# Inheritance

## Improving Structure with Inheritance

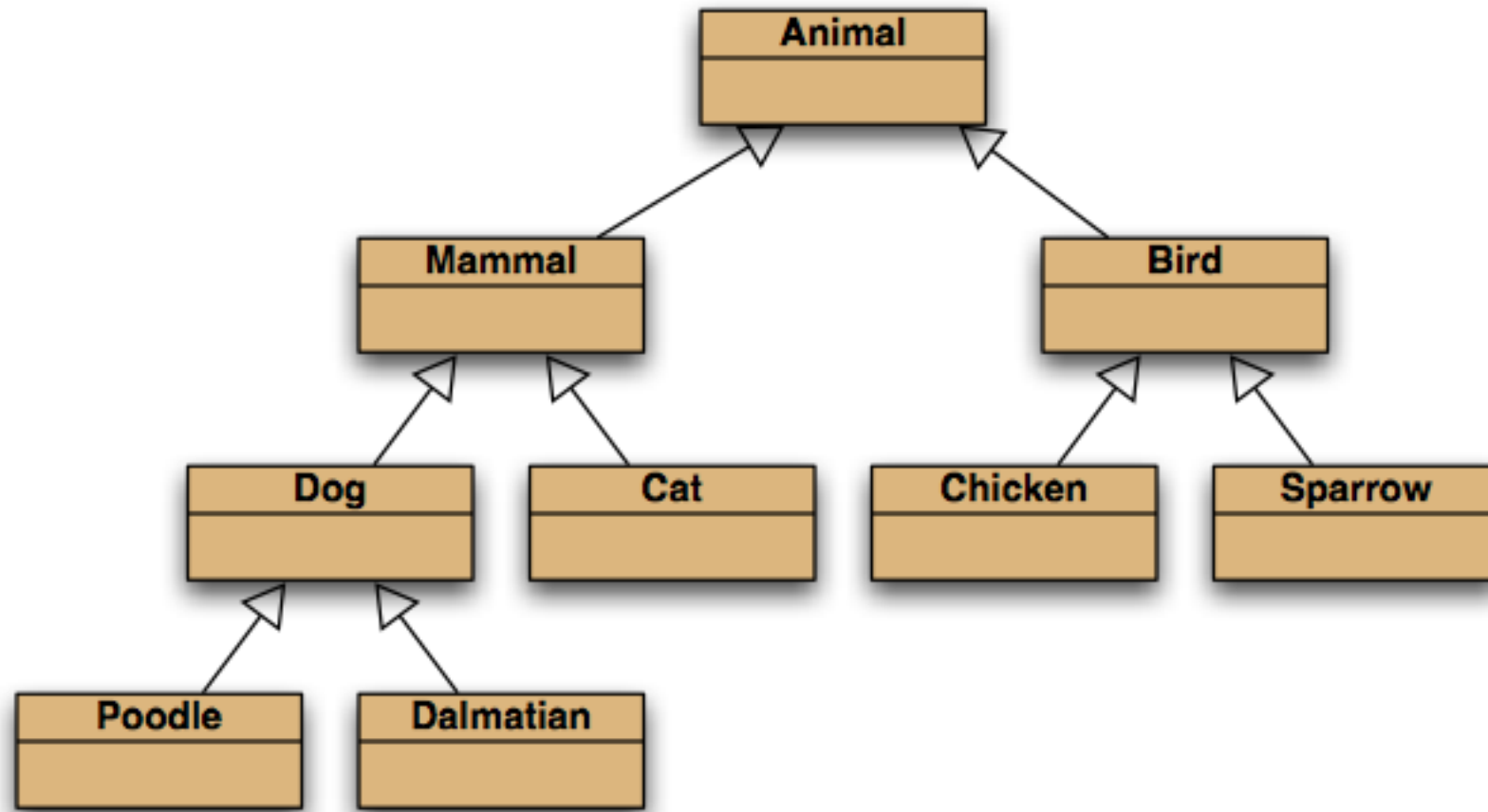
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# Introduction to Inheritance in Java

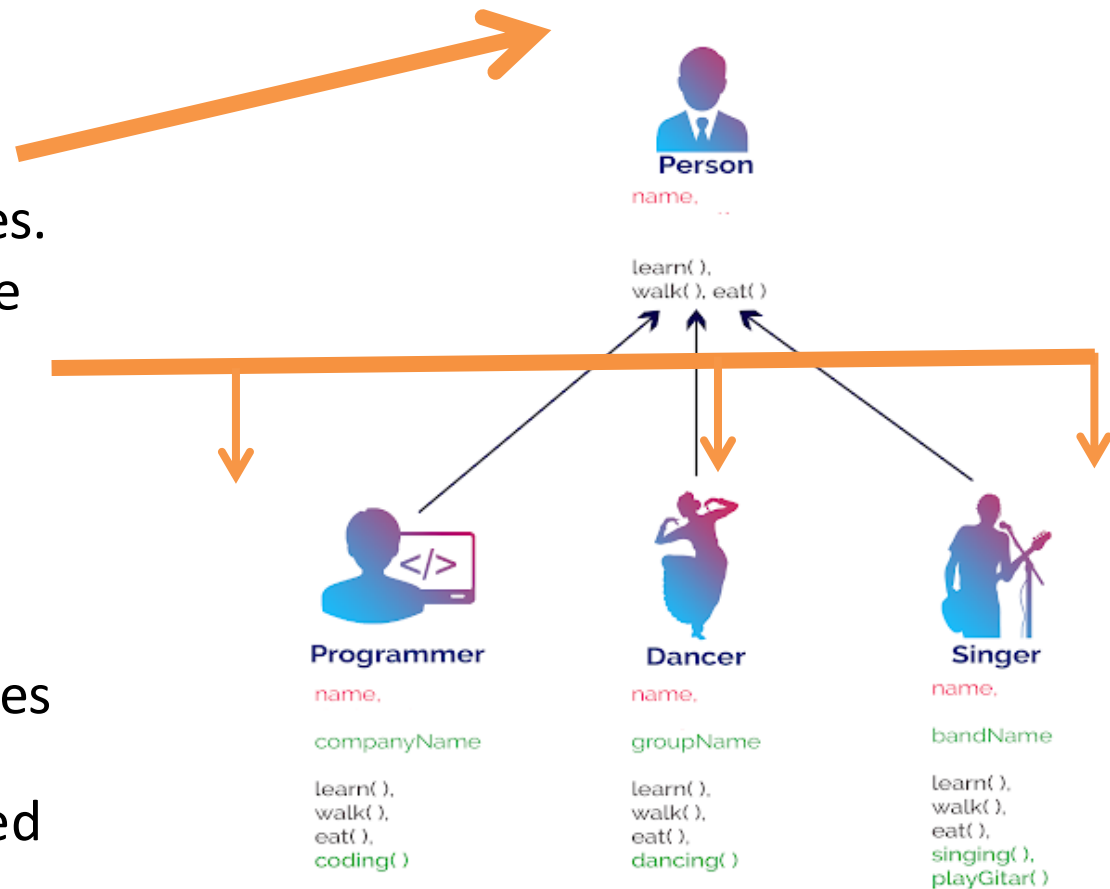
- Inheritance is a fundamental feature of object-oriented programming (OOP) that allows a class to inherit properties and behaviours from another class.
- It promotes code reusability, extensibility, and organisation by allowing new classes to be built upon existing classes

# Inheritance hierarchies



# Basic Concept

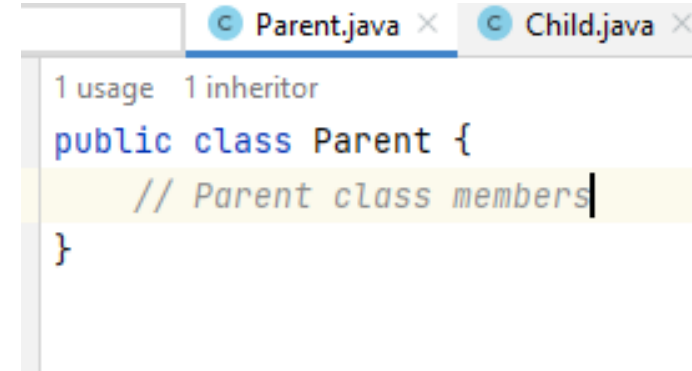
- **Parent Class (Superclass/Base Class):**
  - Defines common properties and behaviours shared by multiple classes.
  - Also referred to as superclass or base class.
- **Child Class (Subclass/Derived Class):**
  - Inherits properties and behaviours from the parent class.
  - Can have its own additional properties and behaviours.
  - Also referred to as subclass or derived class.



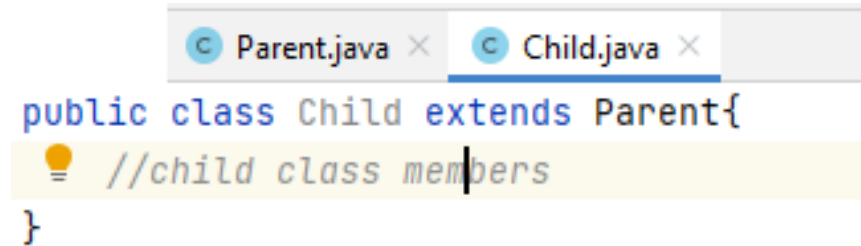
# Syntax for Inheritance in Java

## extends

Indicates that a class is inheriting from another class



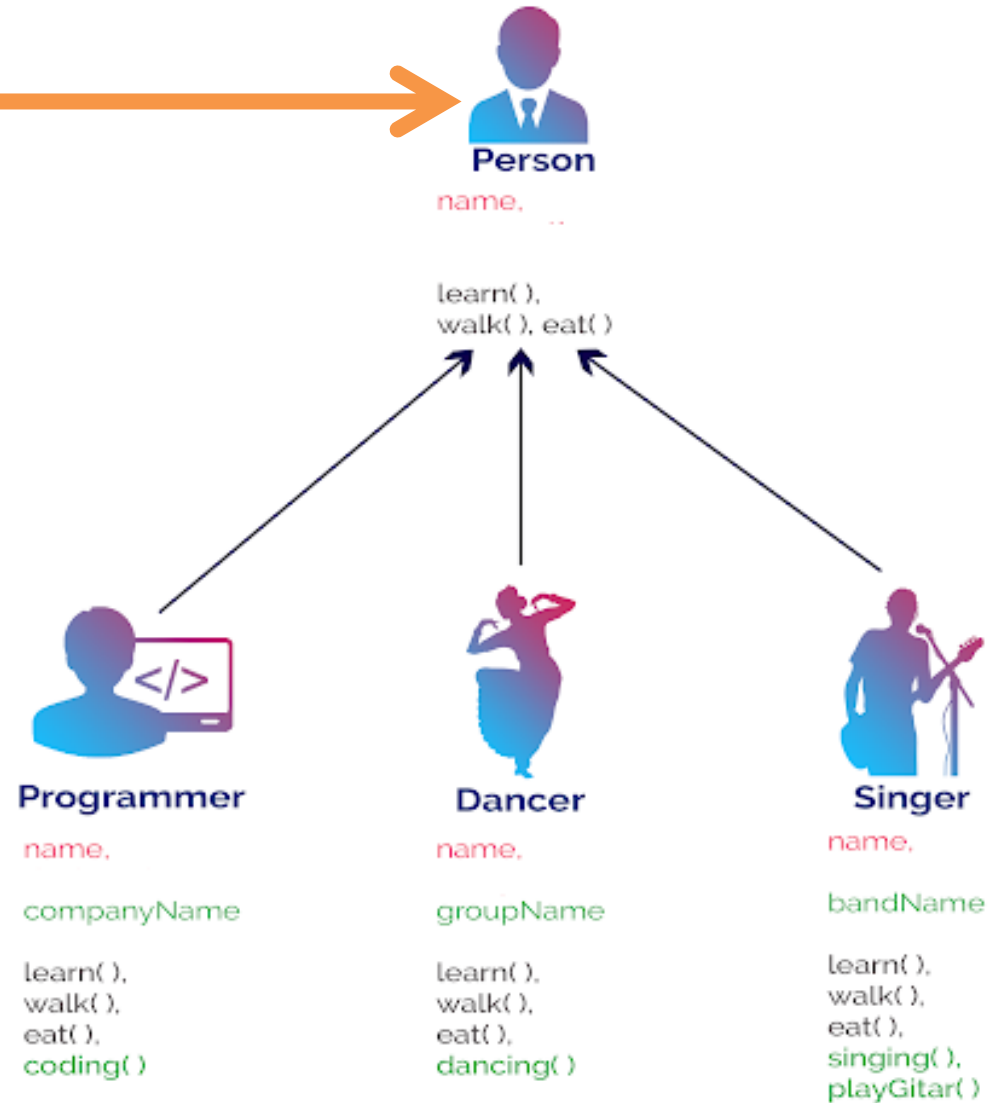
```
1 usage 1 inheritor
public class Parent {
    // Parent class members
}
```



```
public class Child extends Parent{
    //child class members
}
```

# Example - Person

```
public class Person {  
    2 usages  
    String name;  
  
    public Person(String name) {  
        this.name = name;  
    }  
  
    //setters and getters  
    public void learn(){  
        System.out.println("Learns loads");  
    }  
    public void eat(){  
        System.out.println("Eats too much");  
    }  
    public void walk(){  
        System.out.println("Probably doesn't walk enough");  
    }  
    @Override  
    public String toString() {  
        return "Person{" +  
            "name='" + name + '\'' +  
            '}';  
    }  
}
```

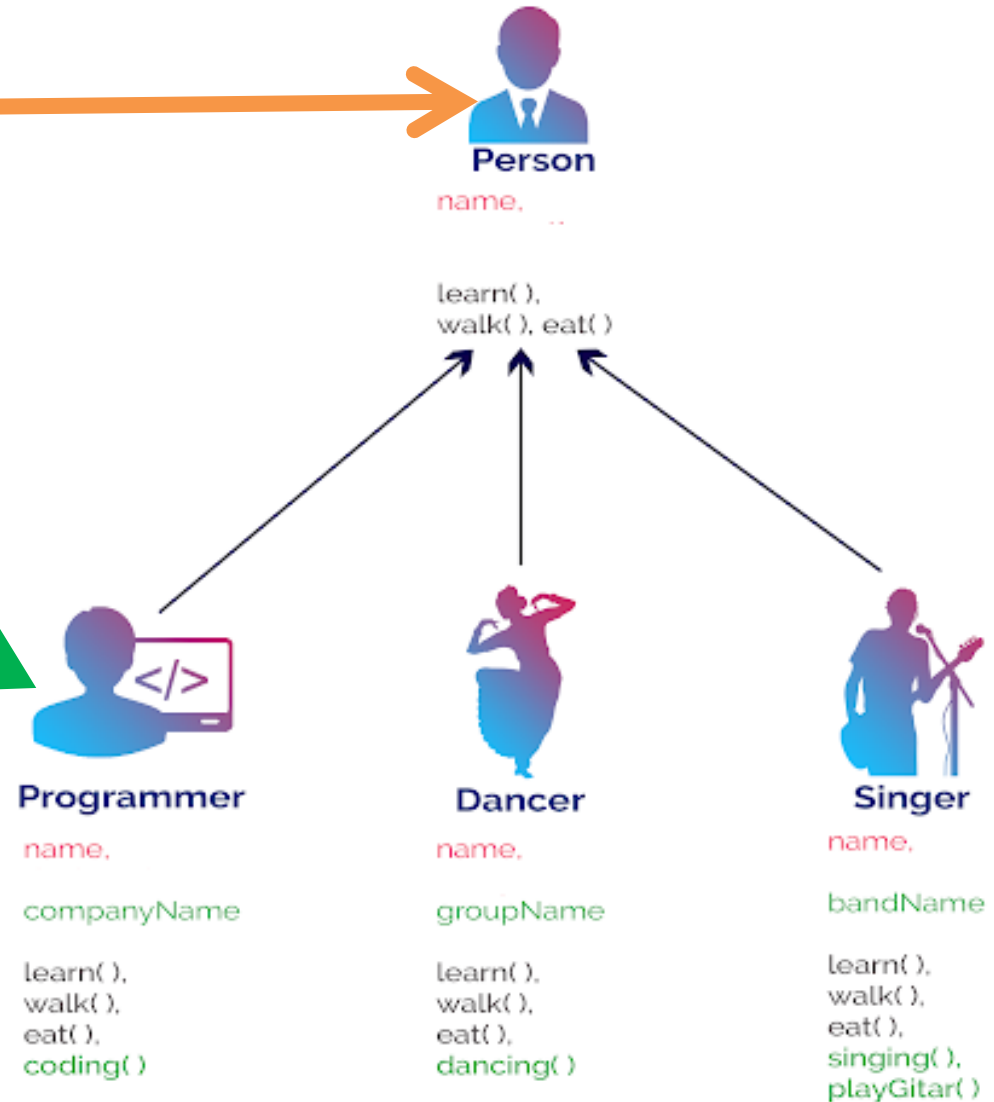


# Example - Programmer

```
public class Programmer extends Person{
    2 usages
    String companyName;

    public Programmer(String name, String companyName) {
        super(name);
        this.companyName = companyName;
    }
    //setters getters
    public void coding(){
        System.out.println("Who doesn't love to be able to code??");
    }

    @Override
    public String toString() {
        return super.toString() + "Programmer{" +
            "companyName='" + companyName + '\'' +
        '}'
    }
}
```



# Example - Dancer

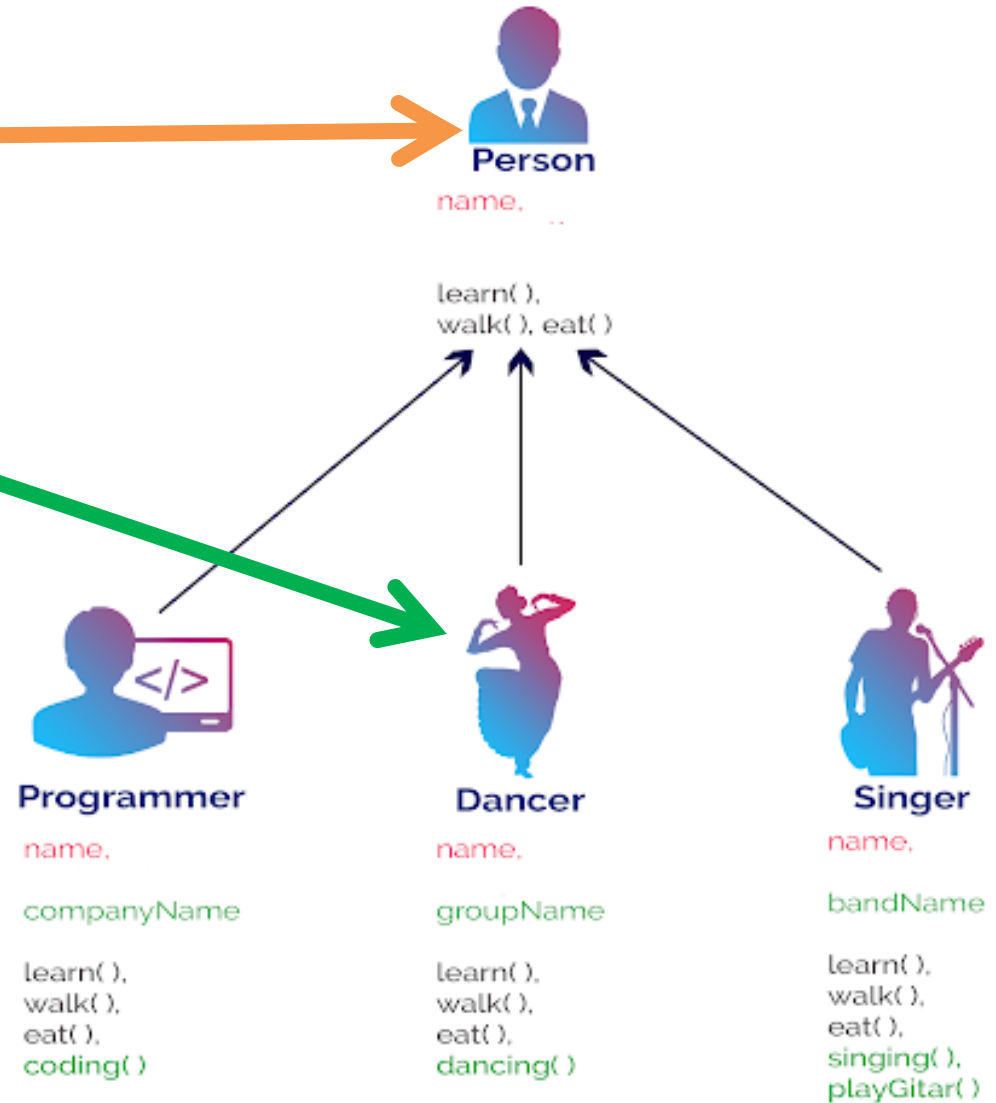
```
public class Dancer extends Person{
    2 usages
    String groupName;

    public Dancer(String name, String groupName) {
        super(name);
        this.groupName = groupName;
    }

    //setters getters
    public void dancing(){
        System.out.println("I love to dance!!!!");
    }

    @Override
    public String toString() {
        return super.toString() + "Dancer{" +
            "groupName='" + groupName + '\'' +
        }

    }
}
```





# Example - Singer

```
public class Singer extends Person{
```

2 usages

```
String bandName;
```

```
} public Singer(String name, String bandName) {  
    super(name);  
    this.bandName = bandName;  
}
```

```
//setters getters
```

```
} public void singing(){  
    System.out.println("I love to sing!!!!");  
}
```

```
} public void playGuitar(){  
    System.out.println("Only when I sing");  
}
```

```
@Override
```

```
} public String toString() {  
    return "Singer{" +  
        "bandName='" + bandName + '\'' +  
        ", name='" + name + '\'' +  
        '}';  
}
```

```
}  
}
```



**Person**

name, ..

learn( ),  
walk( ), eat( )



**Programmer**

name,

companyName

learn( ),  
walk( ),  
eat( ),  
coding( )



**Dancer**

name,

groupName

learn( ),  
walk( ),  
eat( ),  
dancing( )



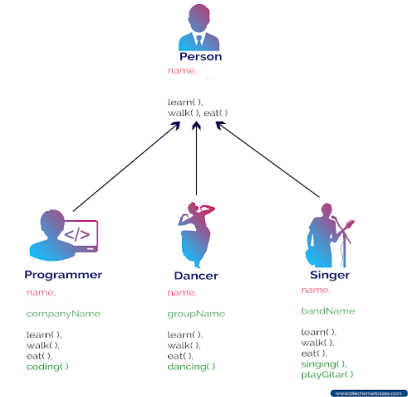
**Singer**

name,

bandName

learn( ),  
walk( ),  
eat( ),  
singing( ),  
playGitar( )

# Example - Driver



```
public class Driver {
    public static void main(String[] args){
        Programmer progObject = new Programmer( name: "Joe Doe", companyName: "Apple");
        Singer singerObject = new Singer( name: "Debbie Harry", bandName: "Blondie");
        Dancer dancerObject = new Dancer( name: "Anna Pavlova", groupName: "Imperial Russian Ballet");

        System.out.println(progObject.toString());
        System.out.println("What can the dancer do?" );
        dancerObject.eat();
        dancerObject.dancing();
        System.out.println("How about the Singer?" );
        singerObject.walk();
        singerObject.playGuitar();
    }
}
```

```
Person{name='Joe Doe'}Programmer{companyName='Apple'}
What can the dancer do?
Eats too much
I love to dance!!!!
How about the Singer?
Probably doesn't walk enough
Only when I sing
```

**Any  
Questions?**

